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% Problem 5.27
% Design lead compensator Dc(s) = k * (s+z) / (s+p), G(s) = 1/(s(s+2))
% Requirement: s = -1 +- j
% ess < 0.2

clear; clc;
wn = sqrt(2);
zeta = 1/sqrt(2); % 0.707
s = tf('s');
G = 1/(s*(s+2));
figure(1);

% % Step 2 - Plot uncompensated root locus
rlocus(G);
title('Uncompensated Root Locus');

% % Step 3 - can't meet specs with uncompensated system, so pick starting
% % values for lead compensator

% z_lead = wn/4; % Rule of thumb: initially 25% of natural frequency.
% p_lead = 10*z_lead; % Rule of thumb: initially placing pole at 10*z;
% D_lead = (s + z_lead)/(s + p_lead);

z_lag = wn / 200;
p_lag = z_lag / 5;
D_lag = (s+z_lag)/(s+p_lag);

figure(2);

rlocus(D_lag * G)




% Define the target pole location
s_target = -1 + 1j;

% Compute the root locus for a fine vector of gains
k_vec = linspace(0, 20, 50000);
[r, k] = rlocus(D_lag * G, k_vec);

% Find the point on the root locus closest to s_target
[min_dist, idx_linear] = min(abs(r(:) - s_target));
[row, col] = ind2sub(size(r), idx_linear);

K_exact = k(col);
achieved_pole = r(row, col);
```

```
% Display the results
fprintf('\n--- Root Locus Verification ---\n');
fprintf('Target Pole:    %.4f + %.4fi\n', real(s_target), imag(s_target));
fprintf('Achieved Pole:  %.4f + %.4fi\n', real(achieved_pole), imag(achieved_pole));
fprintf('Required Gain:  K = %.4f\n', K_exact);
fprintf('Distance Error: %.6f\n', min_dist);

if min_dist < 0.05
    fprintf('SUCCESS: The root locus passes through s = -1 + j!\n');
else
    fprintf('WARNING: The root locus misses s = -1 + j (Error too large).\n');
end

% % Step 5 - Select K value
K = K_exact;


% Step 6 - Plot step response with full lead compensator. Ensure
% specifications are met.
% sys = feedback(K*D_lag*G, 1);
% figure(3);
% step(sys)
%
% % Step 7 - analyze steady state error
% Tau_ten = 50/p_lag;
% t=0:0.001:Tau_ten;
% % Apply ramp input (u(t) = t)
% y=lsim(sys, t, t);
% figure(4);
% plot(t,y,'b',t,t,'r')
% legend("Response", "Ref input")

ess = (2 * p_lag) / (K * z_lag);
fprintf('ess = %.4f\n', ess)
```